



TAMPINES MERIDIAN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

H2 BIOLOGY

Paper 2 Structured Questions

9744/02

14 September 2022
2 hours

SUGGESTED ANSWERS



QUESTION 1

Fig. 1.1 is an electron micrograph of a mammalian liver cell undergoing G1 phase of the mitotic cell cycle. The liver cell contains an abundance of mitochondria and some storage molecules.

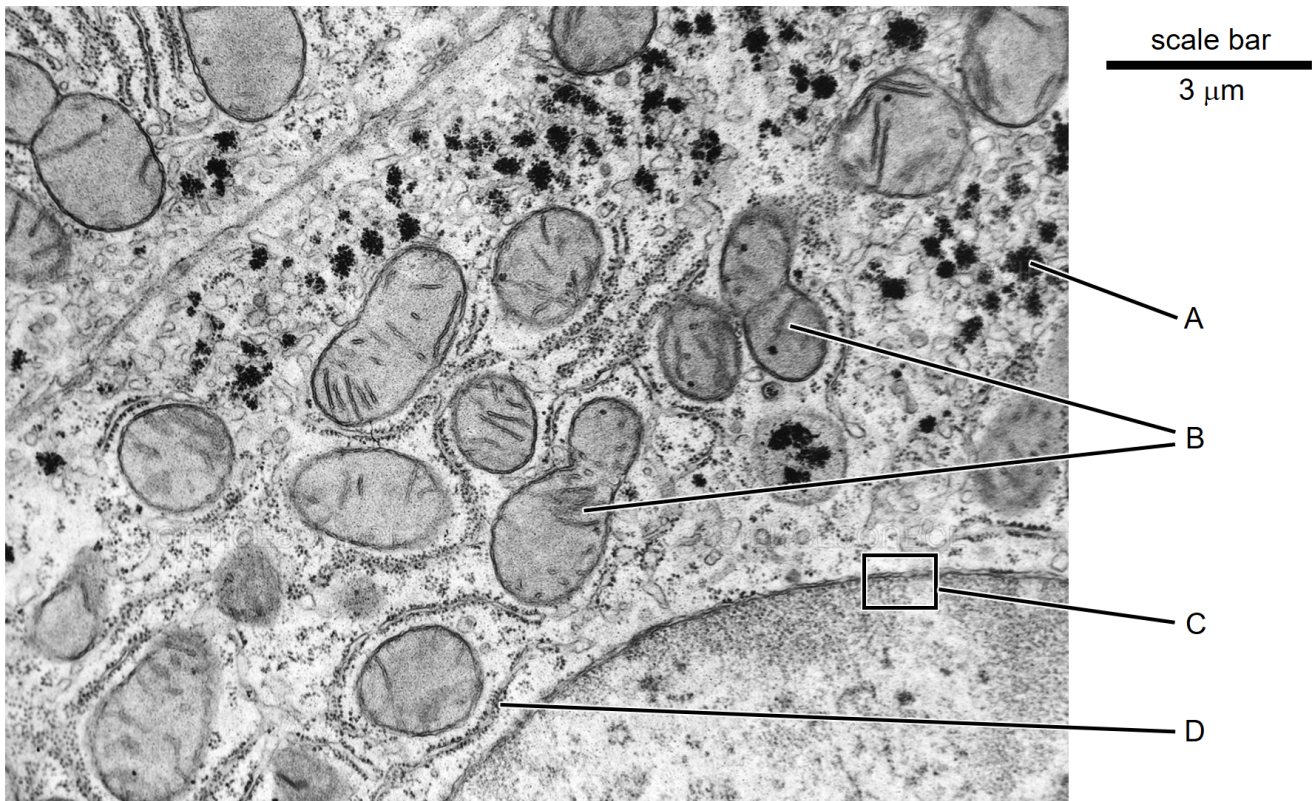


Fig. 1.1

(a) **A** is a storage polysaccharide while **B**, **C** and **D** are organelles or part of an organelle.

Name the storage polysaccharide **A** and organelle **D**. [2]

polysaccharide **A** **glycogen**

organelle **D** **rough endoplasmic reticulum**

(b) Explain why a typical light microscope in a Science laboratory could not have produced the image in Fig. 1.1. [1]

- **Idea that** resolution of light microscope not powerful enough to see **mitochondria** / **rough ER** / **glycogen granules**

(c) Calculate the magnification of the electron micrograph. Show your working. [2]

1. Measured length of scale bar = 27mm = **27,000μm** [1]

2. Magnification = measured length ÷ actual length
 = 27,000μm ÷ 3μm
 = **9,000** [1, allow e.c.f.]

Magnification = ×.....

- (d) The structures labelled **B** on Fig. 1.1 are two mitochondria that are about to divide. Mitochondria synthesize ATP to power cellular processes.

Explain the importance of the division of mitochondria for the cell shown in Fig. 1.1 **and** for subsequent generations of the mammalian liver cells. [2]

for the cell shown in Fig. 1.1

1. Replace worn out / damaged mitochondria
2. Require more mitochondria (to produce more ATP) as the cells enlarge in size

for subsequent generations of the mammalian liver cells

3. So that daughter cells of the liver tissues receive sufficient mitochondria as cells divide.

- (e) Within a cell, substances move between the nucleus and the cytosol. The area labelled **C** in Fig. 1.1 shows an area where this movement occurs.

Fig. 1.2 shows a magnified drawing of area **C**.

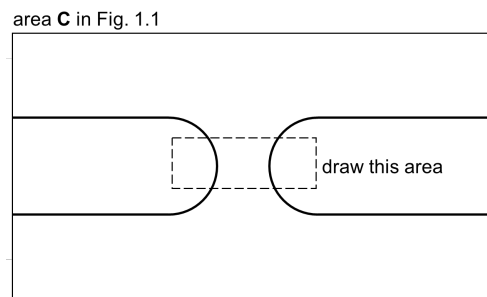
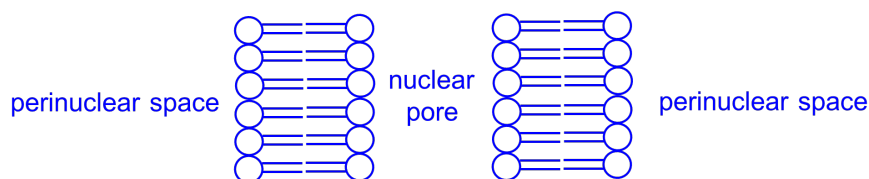


Fig. 1.2

Make a large drawing of the area within the dotted box shown in Fig. 1.2 to show the arrangement of phospholipid molecules. Labeling is **not** required.

You should use the symbol  in your drawing. [2]



- [1] Two membranes with a gap (nuclear pore) in between
 [1] Show each membrane is a bilayer **and** correct orientation of bilayer

[Total: 9]

QUESTION 2

A collagen polypeptide comprises an abundance of the amino acids glycine, proline and hydroxyproline. The repeating tripeptide sequence of glycine–proline–hydroxyproline is very common in collagen.

Fig. 2.1 shows a section of a collagen polypeptide comprising this tripeptide. The R-groups are indicated in bold.

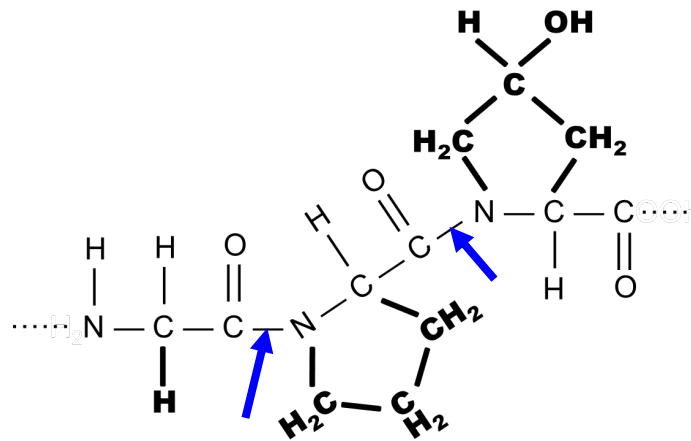


Fig. 2.1

- (a) On Fig. 2.1, use arrows to indicate the peptide bonds that join the three amino acids. [1]
- (b) Unlike glycine and proline, hydroxyproline is **not** directly added to the collagen polypeptide during translation. After translation is completed, no further amino acid is added.

Explain how a collagen polypeptide eventually contains hydroxyproline. [2]

1. post-translational modification

OR

biochemical modification of proteins

2. A hydroxyl group is added to / hydroxylation of the R-group of (selected) proline after the collagen polypeptide is formed

3. Takes place in the rough endoplasmic reticulum / Golgi apparatus

[Any 2]

(c) Fig. 2.2 shows the structure of a single collagen polypeptide. The R-groups of glycine, proline and hydroxyproline are indicated.

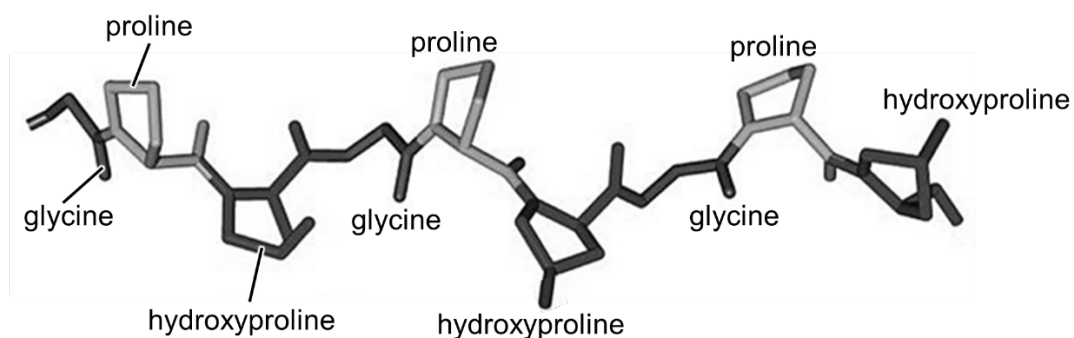


Fig. 2.2

The R-group of glycine is a hydrogen atom. In contrast, proline and hydroxyproline have a rigid and inflexible R-group that tend to cause the polypeptide to bend.

With reference to Fig. 2.2 and the information provided, suggest the role of glycine and that of proline and hydroxyproline in the formation of a tropocollagen molecule. [2]

1. **[glycine]** small R-group allows for tight packing of the three collagen polypeptides
2. **[proline/hydroxyproline]** (both R-groups are rigid thus) drives the twisting of each collagen polypeptide to form a (left-handed) helix

- (d) The enzyme collagenase breaks down collagen. TIMP-1 is a chemical known to inhibit collagenase.

Fig. 2.3 shows the effect of adding a minute amount of TIMP-1 to collagen. The relative concentration of the products of collagen breakdown was recorded every minute for over seven minutes.

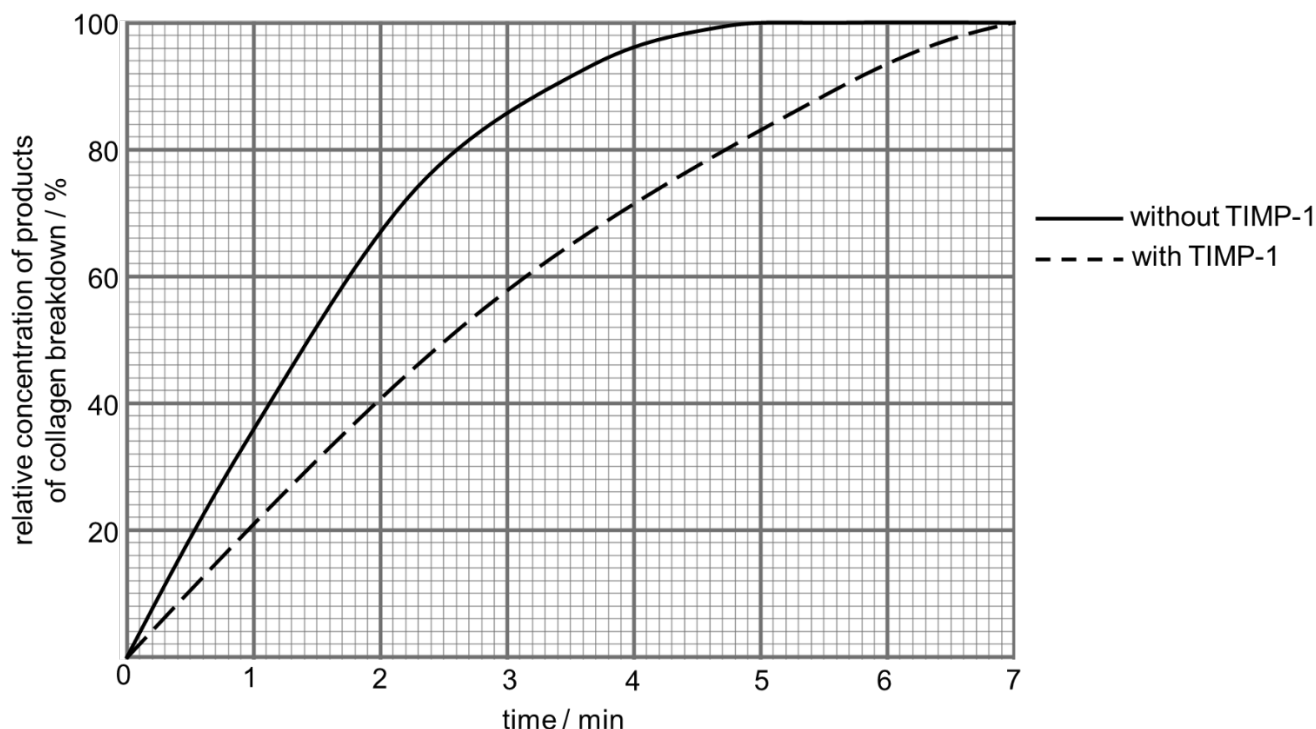


Fig. 2.3

Explain if Fig. 2.3 provides sufficient evidence to conclude if TIMP-1 is a competitive or non-competitive inhibitor of collagenase. [4]

Insufficient + at least 1 from below

- Both types of inhibitors lower the rate of reaction
- Idea that** given enough time, all collagen will still be broken down, as shown in Fig. 2.3
- To determine if TIMP-1 CI or NCI, plot rate of reaction against increasing collagen concentration

[max 2]

- If CI, $V_{\max}$ can be achieved at $[\text{collagen}]_{\text{high}}$...
- ...as collagen outcompete TIMP-1 to bind to the active site of collagenase
- If NCI, $V_{\max}$ cannot be achieved at $[\text{collagen}]_{\text{high}}$
- ...since NCI alters collagenase conformation, effectively lowering no. of functional collagenase

[Any 4]

[Total: 9]

QUESTION 3

Fig. 3.1 represents some of the reactions that take place in a leaf cell of a flowering plant.

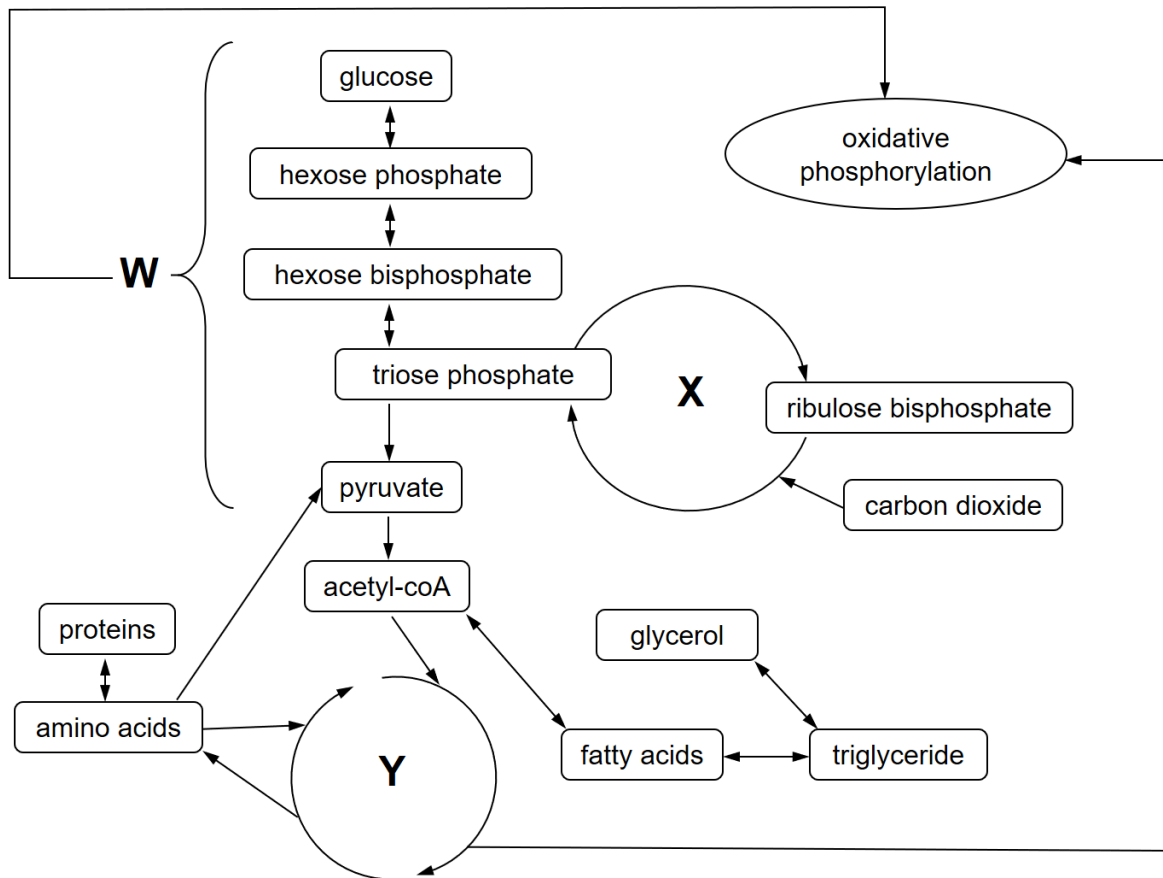


Fig. 3.1

(a) Name the reaction pathways indicated by the letters **W**, **X** and **Y**. [3]

W glycolysis

X Calvin cycle **[Reject: light-independent stage]**

Y Krebs cycle / tricarboxylic acid cycle / citric acid cycle

(b) Explain how the three reaction pathways (**W**, **X** and **Y**) are able to work independently of one another in the same leaf cell. [2]

- Idea of** compartmentalization / membranous organelles to separate reactions
- W** in cytosol, **X** in chloroplast, **Y** in mitochondria
- Ref to** one advantage of compartmentalization: e.g. localizing enzymes

[Any 2]

(c) With reference to the metabolic reactions outlined in Fig. 3.1, state the distinct role of **three named** coenzymes in this leaf cell. [3]

1. NAD / FAD – electron and proton acceptor in glycolysis (NAD only) and Krebs cycle
2. NADP – final electron acceptor in (non-cyclic) photophosphorylation
3. NADH / FADH₂ – donates (high-energy) electrons to the electron transport chain in the inner mitochondrial membrane
4. NADPH – reduces 1,3-bisphosphoglycerate to glyceraldehyde-3-phosphate / triose phosphate in Calvin cycle
5. Coenzyme A – combines with pyruvate to form acetyl-coA

[Any 3]

(d) The carbons in glucose and the carbons in the fatty acids of triglyceride are used to generate acetyl-coA, a two-carbon molecule.

Using your knowledge on biomolecules and the information in Fig. 3.1, explain why **one molecule** of triglyceride can generate more ATP than **one molecule** of glucose. [4]

1. A triglyceride contains three fatty acids
2. Each fatty acid is a long hydrocarbon / contains many carbons, compared to 6 carbons in glucose
3. One glucose generates two acetyl-coA (two-carbon), but one fatty acid/triglyceride generates many acetyl-coA (varies with length of FA)
4. More turns of Krebs cycle with each triglyceride
5. More ATP generated by substrate-level phosphorylation in Krebs cycle
6. More NADH / FADH₂ generated for oxidative phosphorylation to form more ATP

[Any 4]

[Total: 12]

QUESTION 4

A number of different proteins are involved in regulating the cell cycle.

To initiate mitosis, cyclin B and CDK1 (a protein kinase) bind to each other to form a complex called the mitosis-promoting factor (MPF). CDK1, when unbound to cyclin B, is inactive.

MPF phosphorylates proteins that are important for the cell to enter prophase. Examples of these proteins are:

- condensins: the **phosphorylated** form binds to chromatin threads to promote condensation into chromosomes
- lamins: the **unphosphorylated** form binds complementarily to the underside of the nuclear membrane to provide structural support to the nucleus

(a) Explain how phosphorylation of **lamins** by MPF allows the cell to **complete prophase**. [2]

1. Phosphorylation changes conformation of lamins

OR

Phosphorylated lamins no longer complementary in shape to underside of nuclear membrane

2. **Idea of** leads to its detachment/dissociation from the underside of nuclear membrane

3. **[COMPULSORY]** **Idea of** nuclear envelope disintegrates, which hence completes prophase

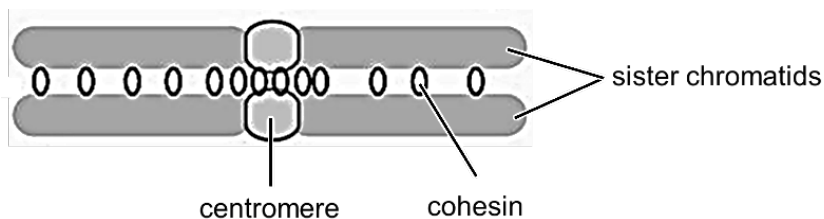
[Any 2]

The condensed chromosomes visible in prophase each consists of sister chromatids, which are held together along their length, including the centromere, by proteins known as cohesin.

Cohesin can be broken down by the enzyme separase.

Separase is initially synthesised in an inactive form by binding to another protein called securin, which physically blocks the active site of separase.

(b) Draw a **labelled** diagram to show how sister chromatids are held together. [3]



two chromatids with centromere [1]
 at least 1 cohesin on centromere and both arms [1]
 correctly label sister chromatids, centromere and cohesin [1]

- (c) Later in mitosis, MPF activates a protein complex called the anaphase-promoting complex (APC).

One role of the activated APC is to catalyse the transfer of ubiquitin molecules to securin and cyclin B. Ubiquitinated proteins are bound for degradation in the proteasome.

Using **all** the information given in this question thus far, explain how

- (i) degradation of securin triggers the cell to enter anaphase. [3]

1. active site of separase no longer blocked by securin
2. separase binds to and breaks down cohesin
3. sister chromatids pulled apart to opposite poles (anaphase)

- (ii) degradation of cyclin B triggers the cell to enter telophase. [3]

1. absence of cyclin B to bind to CDK1
2. CDK1 remains inactive / no MPF

EITHER

3. Condensins become dephosphorylated
4. Chromosome de-condense (telophase)

OR

3. Lamins become dephosphorylated (thus bind underside of nucleus to provide support)
4. nuclear envelope re-forms (telophase)

[Any 3]

- (d) Unlike mitosis, meiosis gives rise to genetically different cells.

Outline **two** ways by which meiosis can give rise to genetically different cells. [2]

1. Crossing over between non-sister chromatids of homologous chromosomes during prophase I
2. Independent assortment and segregation of homologous chromosomes during metaphase I and anaphase I
3. Independent assortment and segregation of chromatids during metaphase II and anaphase II

[Total: 13]

QUESTION 5

DNA methylation usually occurs at the cytosine bases of eukaryotic DNA, catalyzed by DNA methyltransferases (DMT). The methylated cytosine residues are usually immediately adjacent to a guanine nucleotide, resulting in two methylated cytosine residues sitting diagonally to each other on opposing DNA strands.

Different variants of the DMT act in different ways. Fig. 5.1 shows the role of two different variants of DMT (DMT3 and DMT1) at different junctures of DNA methylation.

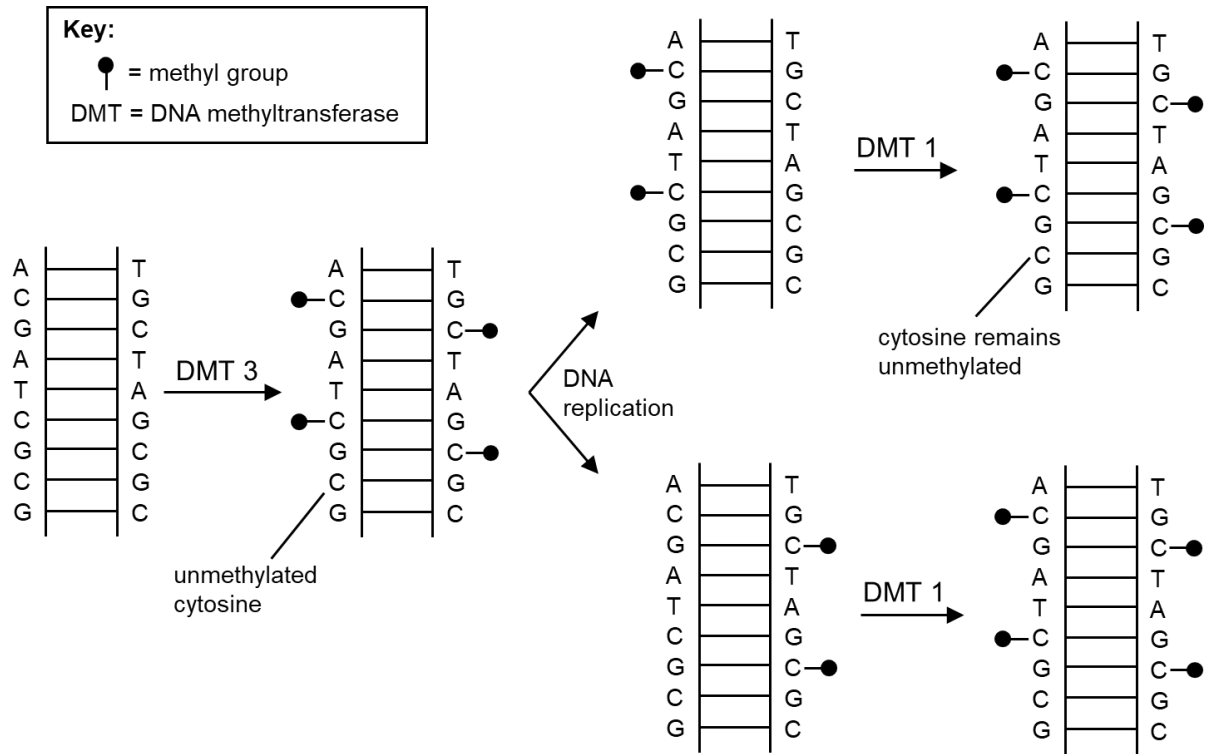


Fig. 5.1

(a) State the name of the enzyme that uses the base sequence of one DNA strand to form the complementary strand. [1]

- DNA polymerase (III)

(b) With reference to Fig. 5.1, state the role of DMT3 and DMT1 in DNA methylation. [2]

1. **Idea that** DMT3 adds methyl groups on certain/some/selected cytosine on both strands
2. **Idea that** DMT1 copies the methylation pattern from an existing DNA strand to the complementary strand after replication

[Reject: DMT3 methylate both strands, while DMT1 methylate one strand]

The *APC* gene was implicated as a possible cause of colon cancer. Samples taken from colon cancer cells of 10 patients and colon cells of 10 healthy individuals were analyzed for their methylation patterns.

Fig. 5.2 shows the DNA methylation patterns on the promoter of the *APC* gene from the 20 samples. The numbers (32–269) represent the base position on the promoter.

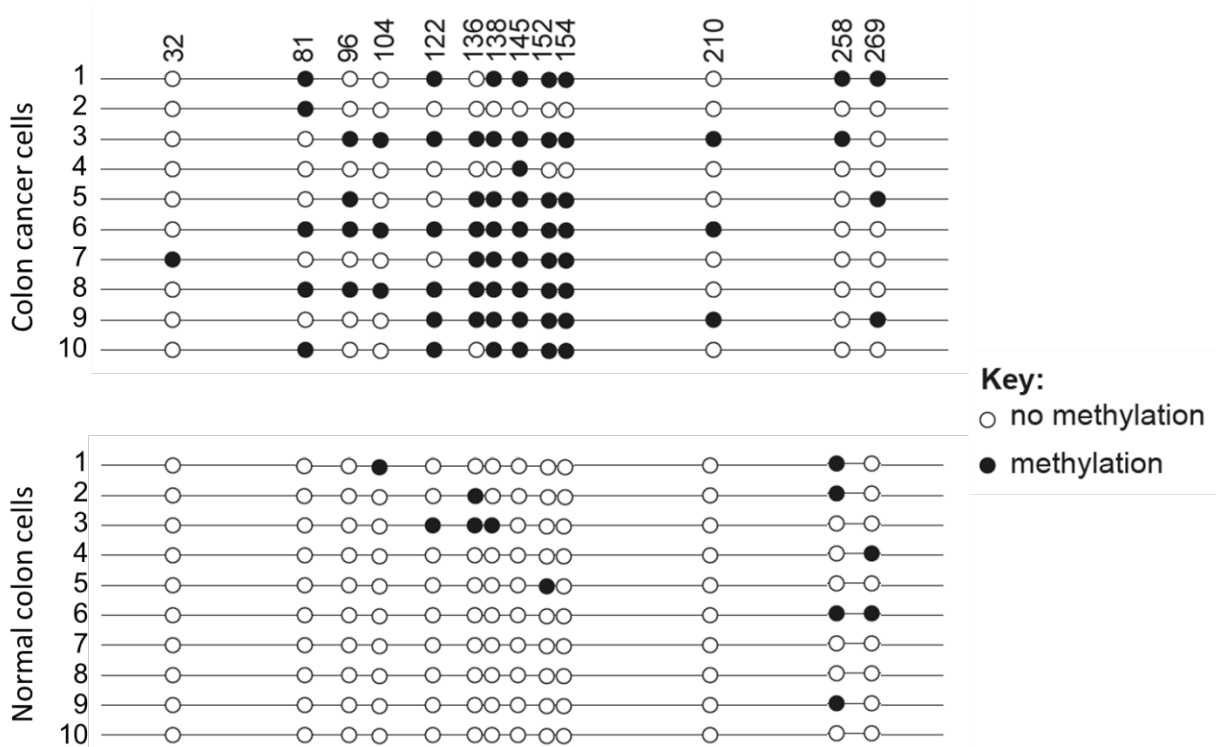


Fig. 5.2

(c) Deduce if the *APC* gene is a proto-oncogene or a tumour-suppressor gene. [1]

- tumour-suppressor gene

(d) Contrast the methylation patterns of cancer and normal samples. [2]

Feature	Cancer	normal
Overall extent of methylation	Greater extent	Smaller extent
Preferential methylation of specific bases	Base 138 / 145 / 152 / 154 preferentially methylated across the 10 samples	No preferential methylation
Any unmethylated base across the 10 samples?	No unmethylated base across the 10 samples	Some bases / base 32 / 81 / 96 / 145 / 210 are not methylated in all the 10 samples
Any samples not methylated at all?	At least 1 base is methylated	3 samples (7, 8 & 10) are not methylated
AVP		

[Any 2]

(e) Suggest why some healthy individuals did not develop colon cancer, despite the *APC* gene being methylated. [2]

1. Cancer is a result of accumulation of mutations in a single cell, hence just *APC* gene alone is not sufficient to result in cancer.
2. Loss-of-function mutations need to occur to other tumor suppressor genes
3. Gain-of-function mutations need to occur to proto-oncogenes
4. **Idea that** The extent of methylation is too small to result in silencing of *APC* gene.

[Total: 8]



QUESTION 6

In white clover, *Trifolium repens*, one gene determines the production of a cyanide-forming substrate. Allele **A** produces the cyanide-forming substrate, whilst allele **a** produces no substrate.

A second gene, located on a different chromosome, determines the production of an enzyme that catalyses the release of cyanide from the substrate. Allele **E** produces the enzyme, whilst allele **e** produces no enzyme.

- Clover that has both **A** and **E** alleles releases cyanide rapidly when its leaves are crushed.
- Clover with **A** but not **E** releases cyanide slowly when its leaves are crushed.
- Clover that does not have **A** cannot release cyanide.

In an experiment, a clover that releases cyanide rapidly was self-pollinated. The following numbers of offspring were obtained:

rapid cyanide release	140
slow cyanide release	49
no cyanide release	67

(a) Construct a genetic diagram to explain the above offspring numbers. [4]

Parent phenotype: releases cyanide rapidly x releases cyanide rapidly

Parent genotype: AaEe AaEe [1]

Gametes: (AE) (Ae) (aE) (ae) (AE) (Ae) (aE) (ae) [1]

Offspring genotype & phenotype: [1]

	(AE)	(Ae)	(aE)	(ae)
(AE)	AAEE rapid release	AAEe rapid release	AaEE rapid release	AaEe rapid release
(Ae)	AAEe rapid release	AAee slow release	AaEe rapid release	Aaee slow release
(aE)	AaEE rapid release	AaEe rapid release	aaEE no release	aaEe no release
(ae)	AaEe rapid release	Aaee slow release	aaEe no release	aaee no release

Offspring phenotypic ratio: 9 rapid release : 3 slow release : 4 no release [1]
Which corresponds closely to 140 49 67

(b) A student attempted to perform a chi-squared (χ^2) test on the offspring numbers.

- (i) Using the ratio from part (a), complete the table below to show the **expected** number of each phenotype of the offspring. [1]

phenotype	observed number (O)	expected number (E)
rapid release	140	144
slow release	49	48
no release	67	64

- (ii) Table 6.1 shows part of the χ^2 table.

Table 6.1

Degrees of freedom	Probability								
	0.9	0.8	0.7	0.5	0.2	0.1	0.05	0.02	0.01
1	0.016	0.064	0.15	0.46	1.64	2.71	3.84	5.41	6.64
2	0.21	0.45	0.71	1.39	3.22	4.60	5.99	7.82	9.21
3	0.58	1.00	1.42	2.37	4.64	6.25	7.82	9.84	11.34
4	1.06	1.65	2.20	3.36	5.99	7.78	9.49	11.67	13.28

The calculated χ^2 value is determined to be 0.273.

Explain what can be concluded from the calculated χ^2 value of 0.273. [3]

1. At $df = 2$, the calculated χ^2 value of 0.273 is less than the **critical value of 5.99** at **$p=0.05$**
2. The probability of the difference in observed and expected numbers being due to chance is **between 0.8 and 0.9**, that is, **more than 0.05**.
3. Hence, the difference is **due to chance**, and the observed number of 140:49:67 **conforms to the expected number** of 144:48:64.

[Total: 8]

QUESTION 7

Erythropoietin (EPO) is a glycoprotein hormone produced by the kidney in response to low blood oxygen concentration. EPO binds to its target stem cell to promote the formation of red blood cells.

- (a) State the name of the stem cell to which EPO binds **and** describe the unique features of this stem cell. [3]

1. <u>myeloid</u> stem cells	OR	<u>blood</u> / <u>hematopoietic</u> stem cells
2. <u>multipotent</u>		<u>pluripotent</u>
3. able to differentiate into red blood cells <u>and</u> (some) white blood cells / named WBC / platelets		able to differentiate into all cell types except the extraembryonic membranes
4. <u>unspecialized</u> cells due to the <u>absence of tissue-specific structures</u>		
5. indefinite self-renewal / proliferation / cell division		
6. (re)-activated telomerase activity		

[Any 3]

- (b) Fig. 7.1 shows the cascade of events that occur upon binding of EPO to the EPO receptor on the target stem cell, leading to the activation of the transcriptional activator, GATA1.

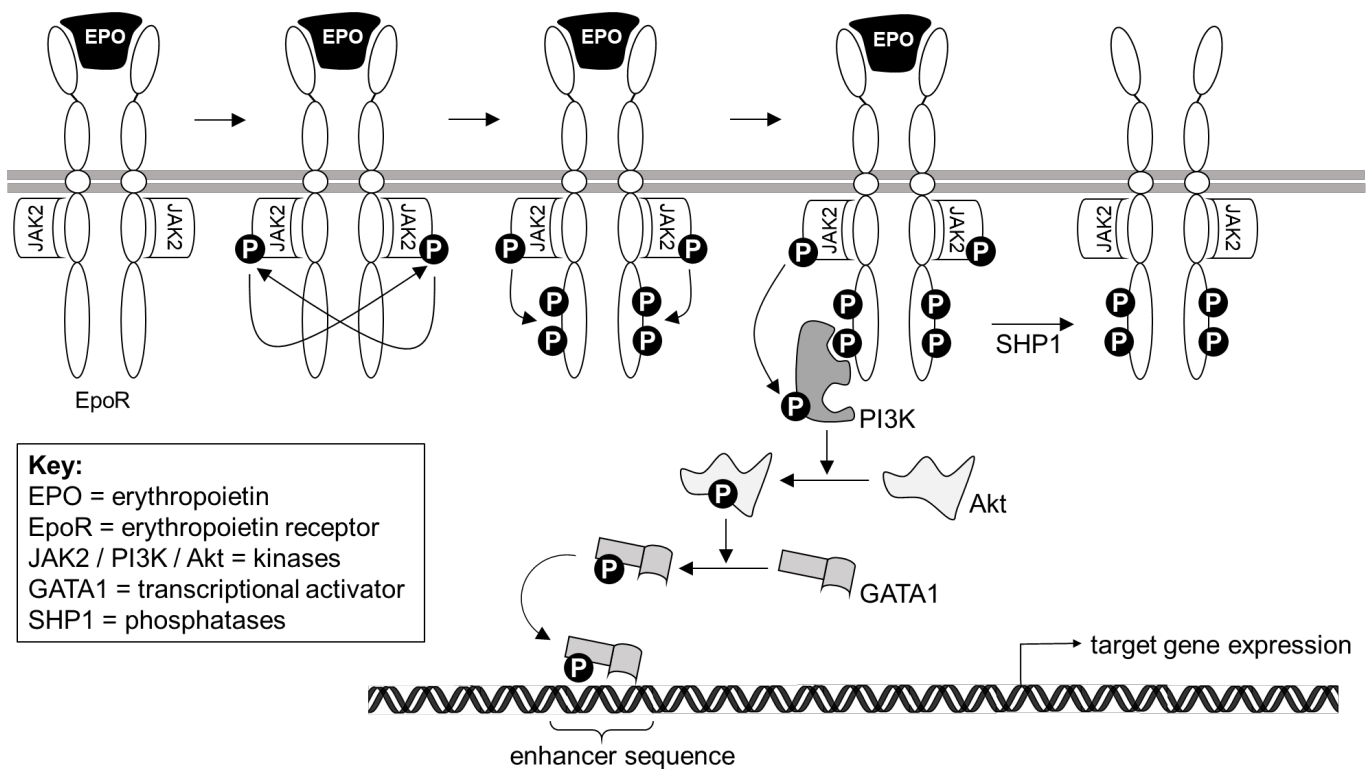


Fig. 7.1

(i) Describe the signal **transduction** events leading to the activation of the transcriptional activator, GATA1. [4]

1. Jak2 kinases cross-phosphorylate and activate each other.
2. Activated Jak2 phosphorylate (multiple tyrosine residues) on the cytoplasmic domain / tyrosine tail of EpoR.
3. PI3K kinase binds to phosphorylated cytoplasmic domain / tyrosine tail of EpoR.
4. Jak2 phosphorylates and activates PI3K.
5. Activated PI3K phosphorylates and activates Akt.
6. Activated Akt phosphorylates and activates GATA1.

[Any 4]

(ii) Explain how the transcriptional activator GATA1 **upregulates** the expression of target genes. [3]

1. Binds to the enhancer sequence
2. *Ref to role of* DNA-bending protein
3. Brings GATA1 in close proximity to the promoter sequence
4. Stabilizes the transcription initiation complex / RNA polymerase
5. Ensures RNA polymerase completes transcription without dissociating prematurely.

[Any 3]

(iii) Name one possible target gene that can be upregulated by GATA1. [1]

- (haemo/ α/β -) globin gene / *any genes involved in mitotic cell cycle* e.g. CDK genes, cyclin genes

(iv) Explain how EPO signaling is terminated. [1]

- SHP1 / phosphatase removes phosphate group from JAK2, inactivating it

(c) As developing red blood cells mature during blood synthesis, membranous organelles such as mitochondria are lost from the cell to make space for haemoglobin molecules. Many organelle-dependent reactions hence do not occur in red blood cells.

(i) Suggest how the loss of mitochondria is a **further** advantage to the proper functioning of red blood cells. [2]

1. Mitochondria use oxygen as the final electron acceptor during aerobic respiration
2. **Idea that** Having lost mitochondria means that red blood cells do not consume any of the oxygen they are transporting, so they can deliver all the oxygen to respiring tissues

(ii) Explain how red blood cells are still able to synthesize a small amount of ATP throughout their lifespan, despite having lost their mitochondria. [2]

1. ATP can be synthesized via glycolysis in the cytosol
2. via substrate level phosphorylation
3. reduction of pyruvate to lactate regenerates NAD for glycolysis to continue.

(iii) Despite the lack of genes due to the absence of a nucleus, new protein molecules continue to be synthesised throughout the lifespan of a red blood cell. However, the rate of protein synthesis decreases with the lifespan of the red blood cell and eventually ceases.

Explain why new proteins can still be synthesised by a red blood cell **and** why protein synthesis eventually ceases. [2]

1. Existing mRNA in the cytosol can still be translated by (80S) ribosomes
2. mRNA degradation / worn-out of ribosomes eventually cease protein synthesis

[Total: 18]

QUESTION 8

Science has identified some two million species of plants, animals and micro-organisms on Earth, but scientists estimate that there are thirteen million more left to discover. The most commonly discovered new species are typically insects, a type of animal with a high degree of biodiversity and accounts for more than half of the species identified. New species are typically discovered in remote places that have not been well studied previously, such as islands.

(a) Suggest why it is impossible to identify all living species on earth. [1]

1. **Ref. to** inaccessible/remote places (e.g. deep oceans, concealed caves)
2. **Idea that** New species are constantly evolving

[Any 1]

(b) Suggest **and** explain why there are so many different species of insects as compared to other animals. [2]

1. **[Suggest]** (Most have) flight ability
2. **[Explain]** Easily dispersed to different environment thus face different selection pressures
3. **[Suggest]** Insects have existed for a long time (before other animals) / evolutionarily older
4. **[Explain]** Hence sufficient time for many rounds of speciation to occur
5. **[Suggest]** Large number of offspring
6. **[Explain]** Affords greater variation, allowing natural selection to act to result in speciation
7. **[Suggest]** Shorter generation time
8. **[Explain]** Within a fixed period of time, more mutations are possible, hence more variation

[Any 1 pair]

(c) Scientists have long been puzzled over how land animals reach remote islands that lie in the midst of vast oceans. Recent studies suggest that animals can travel from one land mass to another by floating on giant rafts of earth and vegetation.

(i) Suggest how climate change may facilitate colonization of remote islands by land animals. [1]

- **Idea of** more frequent natural disasters (like massive landslides) that create land masses for land animals to float on.

(ii) Explain why species on remote islands are usually not found anywhere else on earth. [3]

1. Geographically isolated from mainland
2. No/reduced gene flow from mainland
3. Unique selection pressures
4. Unique set of alleles selected for and against
5. Evolve independently from organisms in the mainland

[Any 3]

(iii) Explain why island populations are much more prone to extinction than mainland populations. [1]

- Low genetic variation (due to Founder's effect), hence more prone to extinction when environment changes.



- (d) Cryptic species are animals that look outwardly similar to another species. Oak Titmouse and Juniper Titmouse, both native to North America, are examples of cryptic species. Until recently, they had been considered as the same species for 151 years.

Fig. 8.1 shows the titmice and their distribution on the North America continent.



Fig. 8.1

- (i) State the **lowest** taxonomic rank on the Linnaean system of classification to which the two populations of titmice share. [1]

- Genus [Reject: *Baeolophus*]

- (ii) Explain why Oak Titmouse and Juniper Titmouse are now classified as two different species. [2]

1. considerable difference in DNA sequences grouped them into two different branches on the phylogenetic tree (phylogenetic species concept)
2. unable to interbreed to produce viable and fertile offspring (biological species concept)
3. both are adapted to different ecological niche (ecological species concept)
4. genetically incompatible and unable to interbreed to produce viable and fertile offspring (genetic species concept)

[Any 2]

- (iii) Suggest **and** explain a **pre-zygotic** isolating mechanism that could prevent successful reproduction between the two populations of titmice in captivity. [2]

Suggest [1]	Explain [1] [Idea that...]
1. Mechanical	reproductive features do not match
2. Behavioral	different courtship rituals / mating calls
3. Gametic	unsuccessful fertilization / gamete incompatibility
4. Temporal	different breeding seasons / becoming fertile at different times

[Reject: any isolation after zygote formation: e.g. reduced hybrid viability]

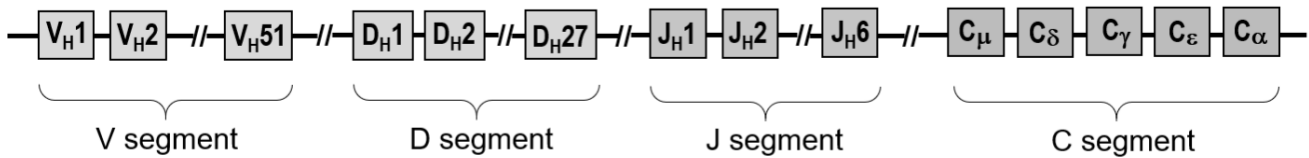
[Any 1]

[Total: 13]

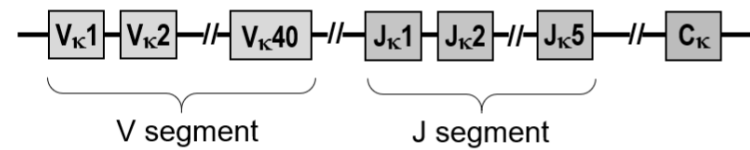
QUESTION 9

In mammals, three genes code for the production of antibodies. Fig. 9.1 shows the structural arrangement of the three genes in humans.

heavy (H) chain gene locus on chromosome 14



kappa (κ) light chain gene locus on chromosome 2



lambda (λ) light chain gene locus on chromosome 22

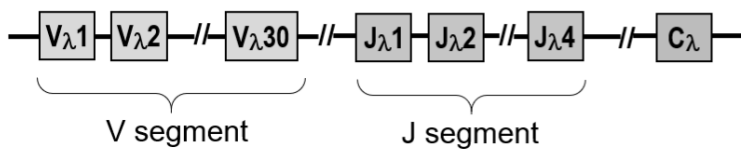


Fig. 9.1

Explain how the structural arrangement of the antibody gene loci creates millions of different antibodies that vary at their antigen-binding site. [5]

1. the genes contain variable/multiple number of V, D (for H chain gene only) and J segments
2. each segment differs in their nucleotide sequence
3. somatic recombination: one random segment from each V(D)J is joined, creating different combinations of V(D)J
4. **cite an example**: H chain recombination creates $V_{10}-D_{24}-J_2$ and $V_{44}-D_2-J_6$ amongst many others / **AVP**
5. remaining segments are removed and degraded
6. expressed / transcribed and translated to form different polypeptides
7. The VDJ segments of H chain gene and VJ segment of L chain gene together encode the antigen-binding site
8. Only either the lambda or the kappa light chain gene is expressed in any B cell
9. Possible number of different antibodies:

$$[(51 \times 27 \times 6) \times (30 \times 4)] + [(51 \times 27 \times 6) \times (40 \times 5)] = 2.64384 \times 10^6$$
[Accept: 2.6×10^6]

[Any 5]

[Total: 5]

QUESTION 10

The deer tick (*Ixodes scapularis*) is an arthropod which sucks blood from humans and other mammals. Some deer ticks can be infected by the bacterium *Borrelia burgdorferi*. When a tick bites a human, the bacterium is often introduced, causing Lyme disease. Lyme disease is a public health problem in North America and, if left untreated, can cause neurological impairment.

Fig. 10.1 represents the two-year life cycle of a tick.

The range of hosts at the three key developmental stages of its life cycle are:

- larva – birds and small mammals
- nymph – humans and other small to large mammals
- adult – humans and other large mammals

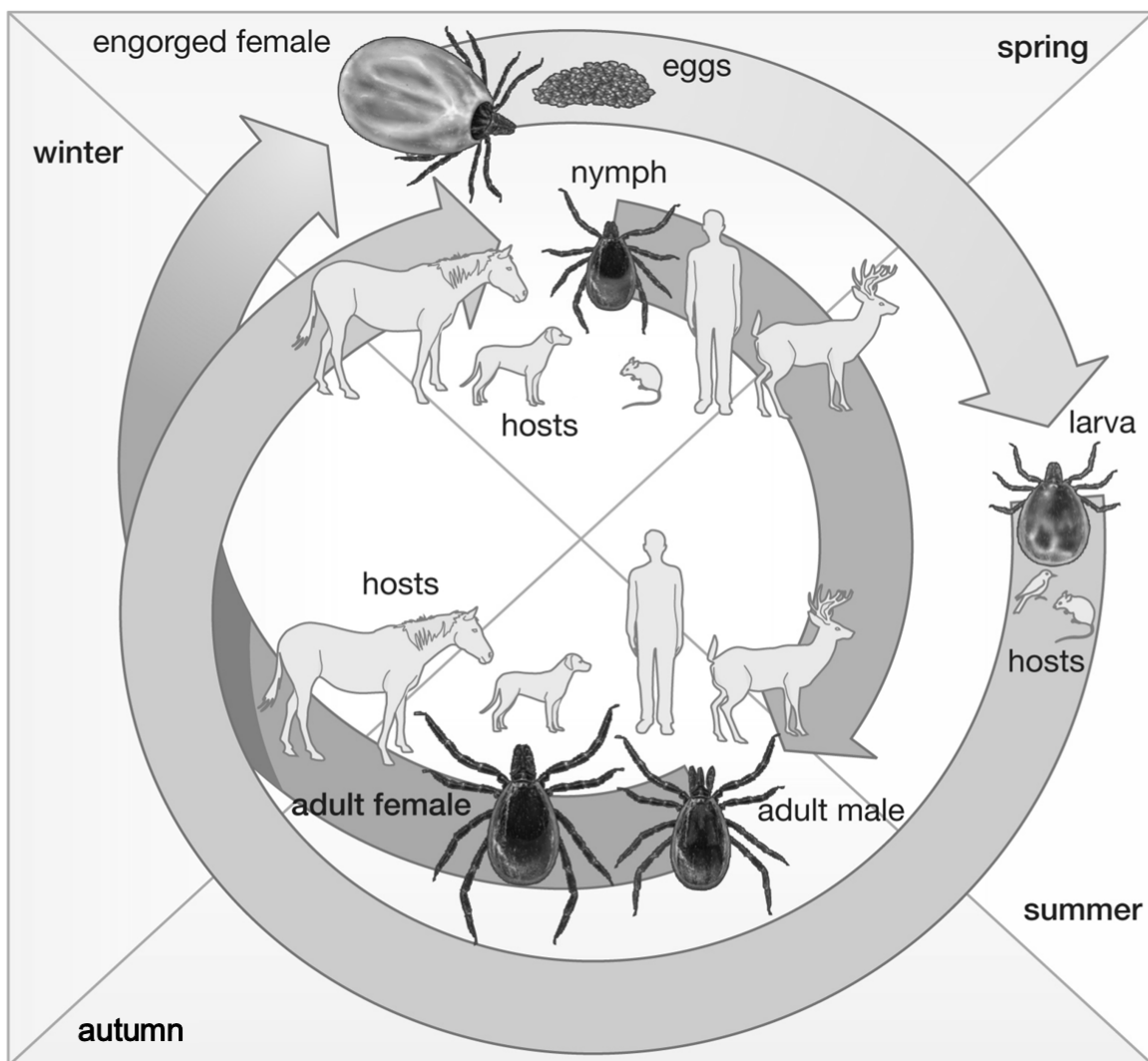


Fig. 10.1

(a) Using the information provided, suggest one possible treatment for Lyme disease. [1]

- **Antibiotics** treatment / **named antibiotics**

Fig. 10.2 shows the developmental stages of deer ticks throughout the four seasons in a densely human-populated area of Canada for the year 2000 and predicted for the year 2080 based on the rate at which the earth is warming.

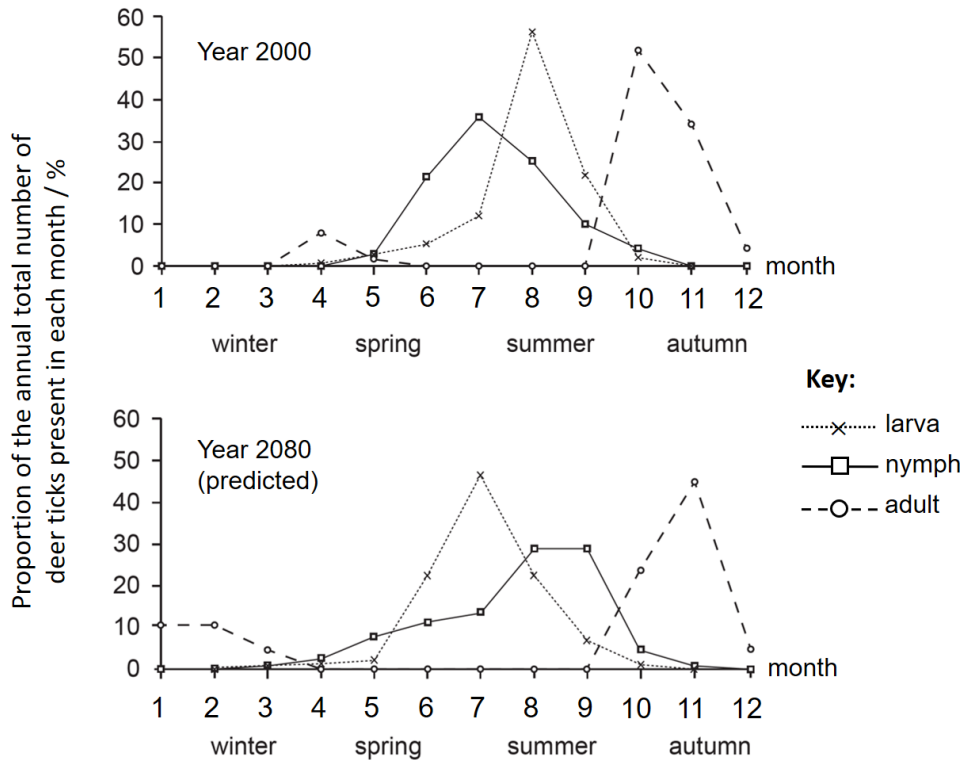


Fig. 10.2

(b) Using Fig. 10.1 and Fig. 10.2, evaluate how global warming will affect the spread of Lyme disease in 2080. [4]

1. Nymphs and adults feed on humans [Fig. 10.1]
2. [data] In Jan, Feb and Mar, adults present at 10%, 10% and 4% respectively in 2080 but absent in 2000...
3. [evaluate]...hence more risk of infection in 2080 in these three months.
4. [data] Adult proportion peaks at 50% in Nov in 2080, vs 43% in Oct in 2000...
5. [evaluate]...hence peak of Lyme disease will shift from Oct to Nov.
6. [data] Nymphs &/or adults are present throughout 2080, but no nymphs and adults from Jan to Mar in 2000...
7. [evaluate]...hence Lyme disease will occur all year round in 2080.
8. Overall, Lyme disease in 2080 is likely to increase / worsen / **AW**
9. **AVP**

[Any 4]

[Total: 5]

☺ END OF PAPER 2 ☺